7 Radium-224 has a half-life of 3.6 days. A freshly prepared sample of Radium-224 has an activity of $1.6 \times 10^{14}$ Bq.

**(a)** Explain what is meant by the term *half-life*.

.....

.....[1]

**(b)** **(i)** Show that the mass of the freshly prepared sample is 27 mg. [2]

**(ii)** A 5.0 mg portion is extracted from the freshly prepared sample. After a period of time, the activity of this portion is found to be $2.1 \times 10^{10}$ Bq.

Determine the time elapsed since the portion was extracted.

time elapsed = ..... days [3]

[Total: 6]